

Lecture 15 [18 September 2026]

Least Squares Value Iteration (LSVI) & Least Squares Policy Iteration (LSPI)

Recap: LSPE algorithm under linear architectures

Linear architectures: Approximate $V_{\omega, \lambda}^{\phi}$ by $B\beta$, $\beta \leftarrow$ returned by LSPE

- Initialise: Specify $\beta \in \mathbb{R}^{I+1}$, $\epsilon > 0$, $\Delta > \epsilon$.
 - Iterate: While $\Delta \geq \epsilon$:

$B = \begin{bmatrix} b_0(s_1) & \dots & b_I(s_1) \\ \vdots & & \vdots \\ b_0(s_m) & \dots & b_I(s_m) \end{bmatrix}$

 - $\underline{V}' \leftarrow B\beta$
 - $\underline{V} \leftarrow L_{\phi} \underline{V}' = r_{\phi} + \lambda P_{\phi} \underline{V}'$
 - $\beta' \leftarrow \Gamma \underline{V}$ $\beta' \leftarrow \underset{\underline{x}}{\operatorname{argmin}} \|\underline{V} - B\underline{x}\|_2^2$
 - $\Delta \leftarrow \|\beta' - \beta\|_2$
 - $\beta \leftarrow \beta'$
 - Terminate: Return β .

The above algorithm approximates $V_{\omega, \lambda}^{\phi}$ for a given decision rule $\phi: \mathcal{S} \rightarrow \mathcal{A}$ using linear architectures.

The following algorithm extends LSPE to approximate $V_{\omega, \lambda}^*$ using linear architectures.

Algorithm 9.3 [Least Squares value iteration, linear architectures]

- Initialise: Specify $\beta \in \mathbb{R}^{I+1}$, $\epsilon > 0$, $\Delta > \epsilon$.
 - Iterate: While $\Delta \geq \epsilon$:

$B = \begin{bmatrix} b_0(s_1) & \dots & b_I(s_1) \\ \vdots & & \vdots \\ b_0(s_m) & \dots & b_I(s_m) \end{bmatrix}$

 - $\underline{V}' \leftarrow B\beta$
 - $\underline{V} \leftarrow L \underline{V}'$
 - $\beta' \leftarrow \Gamma \underline{V}$ $\beta' \leftarrow \underset{\underline{x}}{\operatorname{argmin}} \|\underline{V} - B\underline{x}\|_2^2$
 - $\Delta \leftarrow \|\beta' - \beta\|_2$
 - $\beta \leftarrow \beta'$

- Terminate: Return $\underline{\beta}$

The successive value iterates $\underline{v}_n$ and $\underline{v}_{n+1}$ of LSVI are related as

$$\underline{v}_{n+1} = B \Gamma L \underline{v}_n = T L \underline{v}_n.$$

In some examples, Algorithm 9.3 may not converge.

Moreover, the operator $T L$ may not be a contraction under $\|\cdot\|_2$.

Empirically, LSVI is generally known to perform quite well on many examples.

Algorithm 9.4 [Least Squares Policy Iteration, linear architectures]

1. Initialise: Specify $\phi': \mathcal{S} \rightarrow \mathcal{A}$, $\underline{\beta}$, $\varepsilon > 0$, $\Delta > \varepsilon$.

2. Iterate: While $\Delta \geq \varepsilon$:

a) $\phi \leftarrow \phi'$.

b) Parameter evaluation:

$$\underline{\beta}' \leftarrow (\mathbf{I} - \lambda \Gamma P_{\phi} B)^{-1} \Gamma r_{\phi} \quad (\text{LSPE for } \phi)$$

c) Value function approximation:

$$\underline{v}' \leftarrow B \underline{\beta}'$$

d) Policy improvement:

$$\phi'(s) \leftarrow \arg \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v'(s') \right].$$

e) $\Delta \leftarrow \|\underline{\beta}' - \underline{\beta}\|_2$.

f) $\underline{\beta} \leftarrow \underline{\beta}'$.

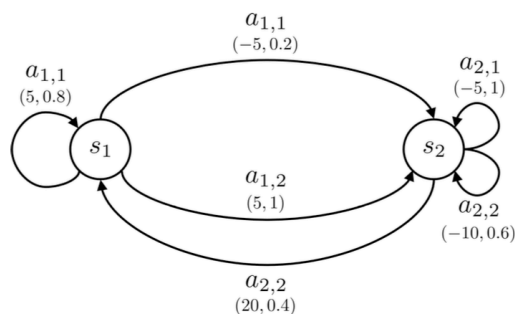
3. Terminate:

Return ϕ' and $\underline{\beta}$.

Remarks on the LSPI algorithm:

- Alternative stopping criteria based on $\text{sp}(\underline{v}' - \underline{v})$, where $\underline{v}, \underline{v}'$ are two consecutive iterates, or based on $\phi'(s) = \phi(s)$ for all $s \in \mathcal{S}$ (with the additional line "choose $\phi'(s) = \phi(s)$ if $\phi(s)$ is an element of $\arg \max$ " added to step 2(d)) may be used.

- An example Where LSPI does not Converge:



$\lambda = 0.9$
 (The values in brackets are $r(s,a,s')$ and $P(s'|s,a)$)

Consider a single basis function $b(\cdot)$ given by

$$b(s_1) = 0.05, \quad b(s_2) = 1.$$

In this case,

$$B = \begin{bmatrix} 0.05 \\ 1 \end{bmatrix}, \quad B\beta = \begin{bmatrix} 0.05\beta \\ \beta \end{bmatrix}, \quad \beta \in \mathbb{R}.$$

The Bellman update step $\underline{v} \leftarrow L\underline{v}' = LB\beta$ in LSVI becomes

$$v(s_1) \leftarrow \max \{ 3 + \lambda(0.8 \times 0.05\beta + 0.2\beta), 5 + \lambda\beta \}$$

$$v(s_2) \leftarrow \max \{ -5 + \lambda\beta, 3 + \lambda(0.4 \times 0.05\beta + 0.6\beta) \}.$$

It can be shown that starting from $\beta_0 \in \{0, -50\}$, LSVI converges quickly and identifies the optimal policy.

However, for both of the above choices of initial β_0 , LSPI cycles and does not converge.

If the basis $b(\cdot)$ is changed or if different β_0 is used, then LSPI and LSVI Converge.

TLDR: It is difficult to predict the behaviour of LSPI/LSVI in general.

- One can apply the bounds from Theorem 5.12 [Lecture 09] to get

$$\begin{aligned} \underline{v}' + \frac{1}{1-\lambda} \min(L\underline{v}' - \underline{v}') \underline{1} &\stackrel{(5)}{\leq} L\underline{v}' + \left(\frac{\lambda}{1-\lambda}\right) \min(L\underline{v}' - \underline{v}') \underline{1} \stackrel{(4)}{\leq} \phi'_{\infty, \lambda} \stackrel{(3)}{\leq} \phi'_{\infty, \lambda}^* \\ &\stackrel{(2)}{\leq} L\underline{v}' + \left(\frac{\lambda}{1-\lambda}\right) \max(L\underline{v}' - \underline{v}') \underline{1} \stackrel{(1)}{\leq} \underline{v}' + \frac{1}{1-\lambda} \max(L\underline{v}' - \underline{v}') \underline{1}, \end{aligned}$$

Where $\underline{v}' = B\beta'$ in LSPI or $\underline{v}' = B\beta$ in LSVI, and ϕ' is as in step 2(d) of LSPI.

The beauty of Theorem 5.12 is that it applies to any $\underline{v}'$, regardless of how it is obtained (via value approximation or otherwise).

- In many examples, the average of quantities appearing after (5) and (2) give a very good approximation for $v_{0,\lambda}^*$:

$$\frac{L\underline{v}' + \frac{\lambda}{1-\lambda} \min(L\underline{v}' - \underline{v}') \underline{1} + L\underline{v}' + \frac{\lambda}{1-\lambda} \max(L\underline{v}' - \underline{v}') \underline{1}}{2} \approx v_{0,\lambda}^*.$$

Algorithm 9.5 [Least Squares Modified Policy Iteration, linear architectures]

1. Initialise: Specify $\bar{M} \in \{0, 1, 2, \dots\}$, $\underline{\beta}$, $\varepsilon > 0$, $\Delta > \varepsilon$.

2. Iterate: While $\Delta \geq \varepsilon$:

a) $\underline{v} \leftarrow B\underline{\beta}$

b) $\underline{v}' \leftarrow L\underline{v}$

$$v'(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') \right]$$

c) $\phi' \leftarrow \operatorname{argmax}_{\phi} L_{\phi} \underline{v}$

d) $m \leftarrow 1$

e) While $m \leq \bar{M}$:

i) $\underline{\beta}' \leftarrow \Gamma \underline{v}'$

ii) $\underline{v}' \leftarrow L_{\phi'} B \underline{\beta}' = r_{\phi'} + \lambda P_{\phi'} B \underline{\beta}'$

iii) $m \leftarrow m+1$

f) $\underline{\beta}' \leftarrow \Gamma \underline{v}'$

g) $\Delta \leftarrow \|\underline{\beta}' - \underline{\beta}\|_2$

h) $\underline{\beta} \leftarrow \underline{\beta}'$

3. Terminate: Return $\underline{\beta}$ and ϕ' .

Remarks:

1. $\bar{M}$ need not be fixed throughout; it can vary from iteration to iteration.

2. If $\bar{M}=0$, $LSMPI = LSVI$; if $\bar{M}=\infty$, $LSMPI = LSPI$.

3. Step 2 e) stops LSPE after $\bar{M}+1$ iterations.

Algorithm 9.6 [LSVI, non-linear architectures]

1. Initialise: Specify $\underline{\beta}$, $\epsilon > 0$, $\Delta > \epsilon$.

2. Iterate: While $\Delta > \epsilon$:

a) $\underline{v}' \leftarrow \underline{f}_{\underline{\beta}}$

$$v'(s) = f_{\underline{\beta}}(s) = f_{\underline{\beta}}(b_0(s), \dots, b_I(s))$$

b) $\underline{v} \leftarrow L\underline{v}'$

c) $\underline{\beta}' \leftarrow \underset{\underline{\beta}}{\operatorname{argmin}} \|\underline{f}_{\underline{\beta}} - \underline{v}\|_2$

d) $\Delta \leftarrow \|\underline{\beta}' - \underline{\beta}\|_2$

e) $\underline{\beta} \leftarrow \underline{\beta}'$

3. Terminate: Return $\underline{\beta}$.

Algorithm 9.7 [LSPI, non-linear architectures]

1. Initialise: Specify $\underline{\beta}$, $\epsilon > 0$, $\Delta > \epsilon$.

2. Iterate: While $\Delta > \epsilon$:

a) $\underline{v}' \leftarrow \underline{f}_{\underline{\beta}}$

b) $\phi'(s) = \underset{a \in \mathcal{A}}{\operatorname{argmax}} \left[r(s, a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s, a) v'(s') \right]$

c) Run LSPE with ϕ' to get $\underline{\beta}'$

d) $\Delta \leftarrow \|\underline{\beta}' - \underline{\beta}\|_2$

e) $\underline{\beta} \leftarrow \underline{\beta}'$

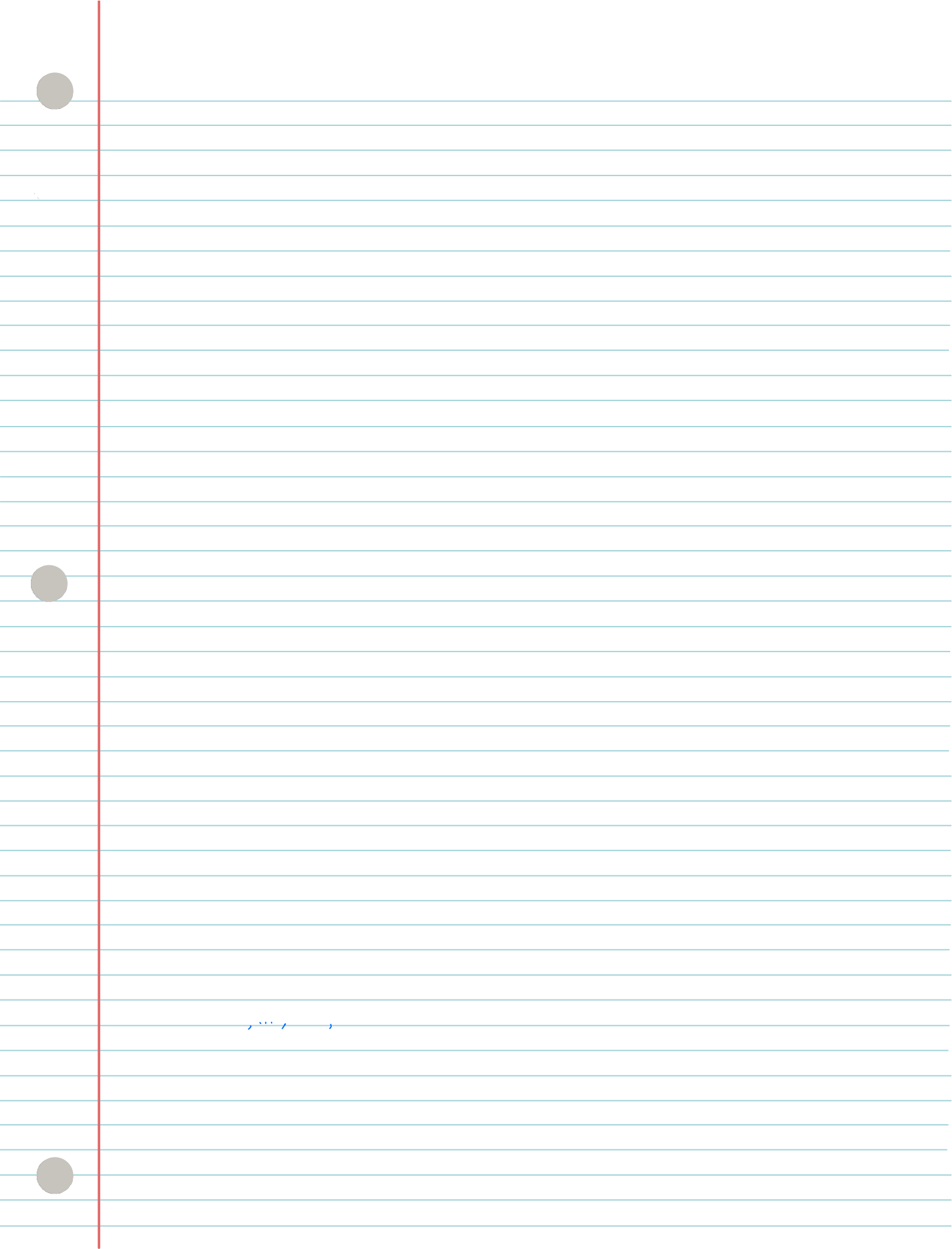
3. Terminate: Return $\underline{\beta}$ and ϕ' .

Remark: In algorithms where ϕ' is not returned (LSVI for linear / non-linear architectures), ϕ' is usually taken to be the greedy decision rule

corresponding to $B_{\hat{p}}$, where $\hat{p}$ is the returned vector from LSVI, i.e.,

$$\phi'(s) = \operatorname{argmax}_{a \in \mathcal{A}} \left[r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a) B_{\hat{p}}(s') \right], \quad s \in \mathcal{S}.$$

- Notice that ϕ is never used in algorithm 9.7, but one can potentially introduce the stopping condition " $\phi'(s) = \phi(s) \forall s$ " to stop (with a corresponding additional line "choose $\phi'(s) = \phi(s)$ whenever $\phi(s)$ is an element of argmax ")
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Infinite Horizon Models: Expected Total Reward Criterion

- Recall

$$V_{\infty, \lambda}^{\pi}(s) = E \left[\sum_{n=1}^{\infty} \lambda^{n-1} r(s_{n-1}, A_{n-1}) \mid S_0 = s \right]$$

← discounted reward criterion

$$= E \left[\sum_{n=1}^{\tau} r(s_{n-1}, A_{n-1}) \mid S_0 = s \right],$$

where $\tau \sim \text{Geo}(1-\lambda)$ is independent of $\{(S_n, A_n)\}_{n \geq 0}$.

- Thus, the discounted reward infinite horizon setting captures problems in which the horizon is a random time that is independent of the MDP states and actions, and Geometrically distributed with parameter $1-\gamma$.

- What if the horizon is determined by the states & actions in an MDP?

- Examples: In Chess, the end-time of the game is dictated by the players' moves

- In gambling, the end point depends on the net worth of the gambler

- In sequential hypothesis testing, the stopping time is a function of the observed trajectory of states and actions.

→ Such problems are best formulated in RL under the "infinite horizon expected total reward" criterion.

Definition 6.1: [Infinite Horizon Expected Total Reward]

For each $s \in \mathcal{S}$ and $\pi \in \Pi^{\text{HR}}$, the infinite horizon expected total reward of policy π , starting from state s , is defined as

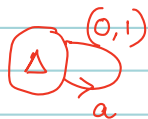
$$V_{\infty}^{\pi}(s) := \lim_{N \rightarrow \infty} E^{\pi} \left[\sum_{n=1}^N R_n \mid S_0 = s \right]$$

$$= \lim_{N \rightarrow \infty} E^{\pi} \left[\sum_{n=1}^N \underbrace{r(S_{n-1}, A_{n-1})}_{r(s,a)} \mid S_0 = s \right]$$

$$r(s,a) := \sum_{s' \in \mathcal{S}} r(s,a,s') P(s'|s,a)$$

The above problem formulation finds applications in :

- Optimal stopping → termination is upon reaching an absorbing state



- Gridworld models
- Sequential hypothesis testing
- Gambling

Important Question: Does the limit in the definition of V_{∞}^{π} always exist? If not, under what conditions does the limit exist? If the limit exists, when is it finite?

Hereafter,
numbering
follows
Chapter
6 of
Chan &
Puterman.

